

ANUPREET SINGH

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EDUCATION

University of Maryland, Baltimore County

(Master of Science in Computer Science, GPA: 3.8/4.0)

Baltimore, MD

Aug 2024 – May 2026

Indian Institute of Information Technology, Sonapat

(Bachelor of Technology in Computer Science and Engineering, GPA: 8.45/10)

Haryana, India

Aug 2019 – May 2023

SKILLS / RELEVANT COURSEWORK

Programming Languages: Python, JavaScript, C++, SQL

Tools and Frameworks: PyTorch, LangChain, Pincecone, pgvector, PostgreSQL, MongoDB, REST(FastAPI, Next.js), React, PostgreSQL, AWS, Redis, Kafka, Docker, Kubernetes, GitHub Actions(CI/CD)

Areas of Interest and Coursework: AI & LLMs, Distributed System Design, Software Engineering (SDLC)

INDUSTRY EXPERIENCE

Data AI Engineer | Platinum Business Services (R&D Team) | Catonsville, MD

Dec 2025 – Present

- Built a priority-ordered fallback in the **RAG chat agent** across 5 embedding models (OpenAI, Anthropic, Ollama), re-embedding and re-running low-citation queries against each successive model's vector index to exploit differences in embedding representation and surface SSP evidence missed by higher-priority indexes, raising evidence-backed control assessment **accuracy from 64% to 92%**.
- Diagnosed an out-of-memory failure in that fallback path, tracing it to freshly pulled 4-5 GB model weights lingering in the OS page cache, and fixed it with targeted `posix_fadvise` eviction, **eliminating model-load failures**.
- Developed **backend services** for an AI-native GRC platform that audits an SSP draft for compliance with NIST 800-53, chunking **~1,200** controls into a per-model vector index with linkage metadata tying each chunk to its parent control and sibling clauses, cutting a control-by-control review from **days of manual work to <10 minutes**.

Software Engineer (AI) Intern | PMA | Tallahassee, FL

Sep 2025 – Dec 2025

- Built an **AI-powered** PRD review platform in React and Django that compares each uploaded draft PDF against a reference PRD and emits schema-validated gap flags for a **human-in-the-loop** reviewer to confirm or dismiss, giving a cohort-wide matrix that replaces a read of every draft and cuts feedback-session prep from **8 to 2 hours**.
- Architected a fully serverless review pipeline on AWS (S3 + CloudFront-hosted React SPA, Django on Lambda, DynamoDB, and an SQS-triggered Lambda AI worker) that retries failed analyses off a dead-letter queue and logs per-report model API spend, sustaining **200+** submissions across 5 cohorts at **~\$3/month** idle cost.

Software Development Engineer | Turfco Landscape Co. | Surrey, BC

Jan 2023 – June 2024

- Architected a self-serve quoting platform that priced jobs with deterministic multi-stage evaluation, JSONB encoded conditions, per-quote audit trails of every rule applied, and a write-invalidated **Redis** cache, all driven by rules editable from an admin console instead of a developer deploy, cutting quote turnaround from **~2 days** to **<1 min**.
- Built the online ordering and booking platform by taking self-serve orders through Stripe Checkout deposits, confirming them with signature-verified idempotent webhooks, and relaying a transactional outbox to **Kafka** workers for order confirmation, customer notification, and crew allocation, taking **\$600K+** in online bookings across **~1,100** orders.
- Designed and deployed the AWS architecture that took the business off a WordPress brochure site, with SPA on CloudFront, autoscaling Express tier behind an ALB across two AZs, Multi-AZ RDS, sized for **~10x** seasonal peaks and load-tested to **~50 quote requests/sec at <120 ms p99** with zero-downtime rolling deploys.

PROJECTS

Nothing Lost in Translation | Lexically constrained Sequence Generation

- Built a **Grid Beam Search decoder** that guarantees positive/negative lexical constraints in any autoregressive sequence generation, demonstrated on English-to-Russian NMT with a **15-point BLEU** gain (25.45 → 40.59).
- Reconstructed a beam search decoder from scratch, outperforming HuggingFace's stock `.generate()` by **1.45 BLEU**.

Bloodborne TAG | Text Adventure Game with natural language command parsing

- Engineered an **LLM Powered command parser** using few-shot Mistral-7B for slot extraction and embedding similarity based intent matching, mapping free-form playtest commands to 14 deterministic actions at **96%** intent accuracy.
- Architected a data-driven game engine that loads a **41-room** world and **60+** items/NPCs from JSON into in-memory session state, gating progression on generic key-item fields so new content ships with **zero** engine code changes.